

Read:

Maxwell's equations time and frequency differential and integral forms	1.1-1.2, 1.7
Continuity condition	1.5.1-1.5.3, 1.7.2
Constitutive relations	1.3
Material parameters	2.2, 2.4.2, 2.4.3, 2.8
Helmholtz equation	3.3, 4.2.1
Plane waves	4.2.2, 4.3
Poynting theorem	1.6, 1.7.3
Polarization	4.4

1. Convert the following time domain expressions for the electric and magnetic field into frequency domain expressions. Show each mathematical step in your calculations. A miraculous answer without the complete calculation will be judged harshly.

- (a) $\mathbf{E}(t) = 4 \cos(\omega t - 28z) \hat{\mathbf{y}}$ V/m
- (b) $\mathbf{E}(t) = 9 \sin(\omega t - 23z) \hat{\mathbf{x}}$ V/m
- (c) $\mathbf{H}(t) = 7 \cos(\omega t - 23z) \hat{\mathbf{y}}$ mA/m
- (d) $\mathbf{E}(t) = 4.6 e^{-48\pi z} \cos(\omega t - 139z + \pi/4) \hat{\mathbf{y}}$ V/m

2. Convert the following frequency domain expressions to the time domain. Show each mathematical step in your calculation.

- (a) $\mathbf{E} = 15 e^{-j23z} \hat{\mathbf{x}}$ V/m
- (b) $\mathbf{E} = 32 e^{-(52+j135)z} \hat{\mathbf{y}}$ V/m
- (c) $\mathbf{H} = [(2+j2)\hat{\mathbf{x}} + (1-j3)\hat{\mathbf{y}}] e^{-j245z}$ A/m
- (d) $\rho = 3 j e^{j\pi/2}$ C/m³

Be sure to place each of the vector components in simplest magnitude and phase form before converting to the time domain.

3. Assuming a homogeneous region where there is no current, charge, or loss:
 - (a) Use Maxwell's equations to convert the frequency domain electric field expression in 2(b) to the corresponding magnetic field. Show all mathematical steps.
 - (b) Use Maxwell's equations to convert the frequency domain magnetic field in 2(c) into a frequency domain electric field into a frequency domain magnetic field. Show all mathematical steps.
4. Use 1(b) and the corresponding electric field in found in problem 1(c) to calculate the time average power density. Show all mathematical steps.
5. Starting with the frequency domain form of Maxwell's equations derive the magnetic field vector Helmholtz equation. Number your Maxwell equations and at every step write briefly which Maxwell equation and/or which identity you are using from the Vector Identities section of Appendix II.3.1 (pg. 958). Assume there is no electric or magnetic current and no electric or magnetic charge.